Overview

The Vela-7 weather station records temperature, humidity, air pressure and wind speed. By default it takes one reading every 10 minutes. It is powered by a 20-watt solar panel that charges a 12 Ah battery.

Installation

Mount the Vela-7 mast 2 metres above the ground. Keep the station at least 3 metres away from walls and trees. Secure the mast with four guy wires. Use the built-in bubble indicator to level the unit.

Calibration

Calibrate the temperature sensor every 12 months in an ice bath at 0 degrees Celsius. The pressure sensor is calibrated at the factory and drifts by at most 0.5 hPa per year. The tipping-bucket rain gauge measures 0.2 mm of rain per tip.

Troubleshooting and data

A blinking red LED means the battery is low. A steady blue LED means the station is uploading data. Hold the reset button for 10 seconds to restart the unit. Readings are stored as CSV files on a microSD card of up to 32 GB.